Experimentation on Visual Question Answering with Abstract Scenes and Multiple Choice Answers

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Abstract

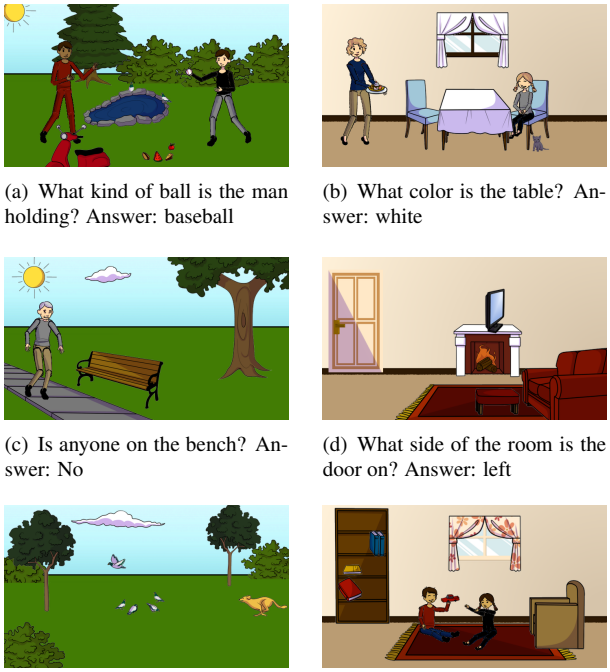
In this project we explore several neural network architectures for the Visual Question Answering task on abstract scenes involving both the language and the visual model. In particular, we examine the effect of fine-tuning the last layers of popular deep convolutional neural networks such as ResNet and AlexNet for obtaining image features. For the language model, we try various ways of organizing the LSTM units in the network and using different non-linearity functions to get word-embedding. We also compare different late fusion strategies of the two models, including addition and side-by-side concatenation to obtain joint vector embedding of the question and image. Finally, we propose a model that outperforms the baseline for around 8% on the VQA validation data for abstract scenes and multiple choice questions.

1. Introduction

A VQA system takes an image and a natural language question about the image as input, and produces a natural language answer as an output. Solving VQA requires attending to multiple sub-problems such as fine-grained recognition, object detection, knowledge-based and commonsense reasoning, which makes it a complicated problem to approach. At the same time, this multi-modal knowledge property, along with the currently existing quantitative evaluation metric, makes it a potential candidate for an "AI-complete" problem.

In this paper, for simplicity, we use abstract scenes instead of natural images. Doing so removes the need for extracting fine details from the image, and allows us to focus on high-level reasoning. In addition, we use only multiple choice questions, as it restricts the space of possible answers and thus makes the task easier. See figure 1.

Our goal for this project is to examine the performance of variants of an existing model intended for the original VQA problem (real images, open-ended questions). Through our examination, we attempt to identify the features that should be replaced or added to this model in order to achieve better accuracy in our simplified VQA setup.



(a) What kind of ball is the man holding? Answer: baseball

(b) What color is the table? Answer: white

(c) Is anyone on the bench? Answer: No

(d) What side of the room is the door on? Answer: left

(e) How many birds are afraid of the dog? Answer: 1

(f) How many books are there? Answer: 4

Figure 1: Examples of abstract scenes along with a question and correct answer. In this project we focus on answering multiple-choice questions in the abstract world

2. Related Work

The task of VQA has gained increasing interest in the AI research community since the emergence of the seminal paper of *Antol et al.* [1] which proposed the VQA task of open-ended questions on real images. Moreover, it introduced the VQA datatset containing 0.25M images, 0.76M

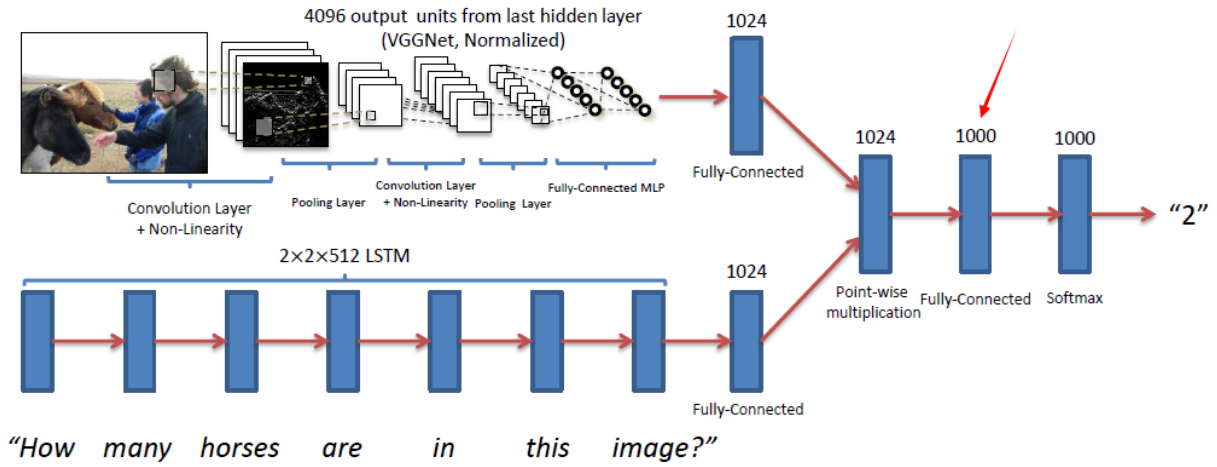


Figure 2: Architecture of the baseline model consisting of image and question channel which are eventually fused together.

Source: VQA: Visual Question Answering *Antol et al.*[1]

questions, and 10M answers for both real and abstract scenes and both open-ended and multiple choice questions, which has become the *de facto* benchmark for the VQA task. It was found that this dataset has a strong language prior, meaning one could obtain good superficial performance by using the question only, ignoring the associated image. This was addressed in [2], where a balanced version of the dataset was created for real images. Similar efforts were made to balance the binary (yes/no) questions for abstract scenes [3]. The dataset for abstract scenes and multiple choice answers, which we use for this project, remained the same in both versions.

The most common approach for solving the VQA task has been using techniques such as joint embedding of image and question features [1]. More recent papers, use attention mechanisms which enable focusing only on the relevant parts of the image and/or question [4] [5]. However, most of these papers consider only the real images. We take an existing VQA model as our base model, adapt it to work with abstract images and compare accuracy by changing one architectural segment at a time. Finally, we try combining different designs together and compare the performances of these approaches.

3. Approach and Experimentation Methods

Now we will describe our dataset, our baseline model and the new methods we experiment with.

3.1. Dataset

VQA abstract dataset consists of 50K abstract scenes, with 3 questions for each scene, with train/val/test splits of 20K/10K/20K scenes respectively. Each question has 10 human-provided ground-truth answers. Questions are categorized into 3 types: yes/no, number, and other. For the multiple-choice task, 18 possible answers are provided. In our project, we use the validation data as a test data, as we do not have access to annotated test data.

3.2. Baseline model

As a starting model we use the best performing model of *Antol et al* [1] (code provided by *Lu et al.* [6]), which achieves an accuracy of 58.16% on open-ended and 63.08% on multiple-choice on test-standard split for **real** images. It consists of two networks: one processes the image (using Convolutional Neural Network (CNN) to extract image features) and the other processes the question (using Long Short-Term Memory (LSTM) recurrent neural network to obtain question embedding). The image and question features obtained from the two networks are fused together and passed through a fully connected (FC) layer to obtain a softmax distribution over the space of answers. See figure 2.

We trained this default model on **abstract scenes** for 75K iterations (the default was 150K) and we got accuracy of **59.28%** (on validation data). We use the standard evaluation scripts available online. In the subsequent parts we will explain what modifications we made in order to achieve higher accuracy.

3.3. Methods

Image: The original model used normalized activations of the last hidden layer of VGGNet-19 [7] to obtain embedding for the images which are then transformed to 1024-dim by a fully-connected layer + tanh non-linearity to match the

LSTM embedding of the question. Instead of VGGNet-19, we try the following CNNs to obtain image embedding:

1. **AlexNet**: use pretrained AlexNet [8] and obtain the 4096-dim output of the last hidden layer.
2. **ResNet-50**: use a pretrained Resnet-50 [9] to get the output of the last hidden layer, and reshape it to get 4096-dim vector.

Question: In the baseline model an LSTM with two hidden layers is used to obtain 2048-dim embedding for the question. The embedding obtained from the LSTM is a concatenation of last cell state and last hidden state representations (each being 512-dim) from each of the two hidden layers of the LSTM. This is followed by a fully-connected layer + tanh non-linearity to transform 2048-dim embedding to 1024-dim. Each question word is encoded with 300-dim embedding by a fully-connected layer + tanh non-linearity which is then fed to the LSTM. The input vocabulary to the embedding layer consists of all the question words seen in the training dataset. We experiment with the following:

1. **4Layers256Units**: increase the number of hidden layers to 4, decrease the number of hidden units to 256.
2. **8Layers128Units**: increase the number of hidden layers to 8, decrease the number of hidden units to 128.
3. **ReLUAct**: use ReLU non-linearity as opposed to Tanh in the word-embedding of the question.
4. **SigmoidAct**: use Sigmoid non-linearity as opposed to Tanh in the word-embedding of the question.

Multimodal fusion: In the original model the image and the question embeddings are combined via multiplication to obtain a single embedding. For the sake of experimentation, we try different fusion methods:

1. **ConcatSideBySide** Concatenate the image and question embedding side by side
2. **ADDforFusion** Add the image and question vector embedding

The resulting joint embedding is then passed to a fully connected neural network classifier with 2 hidden layers and 1000 hidden units (dropout 0.5) in each layer with tanh non-linearity, followed by a softmax layer to obtain a distribution over the possible answers. The entire model is learned end-to-end with a cross-entropy loss, keeping the image weights frozen.

3.4. Combined methods

We try combining different methods from each domain (image, question and multimodal fusion) to investigate whether joining two well performing methods results in better scores:

1. **ResNet50-2x512-Add-Tanh**: use ResNet50 image features, use 2 layers with 512 units each for the LSTM, use Addition as a fusion method and use Tanh as a non-linearity for word embeddings
2. **ResNet50-4x256-Mul-Tanh**: ResNet50 image features, 4 layers with 256 units each for the LSTM, use Multiplication as a fusion method and use Tanh
3. **ResNet50-4x256-Concat-ReLU**: ResNet50 image features, use 4 layers with 256 units each for the LSTM, use Side by Side Concatenation, use ReLU as a non-linearity
4. **ResNet50-8x128-Add-Tanh**: use ResNet50 image features, use 8 layers with 128 units each for the LSTM, use Addition as a fusion method and use Tanh as a non-linearity for word embeddings
5. **VGG19-4x256-Concat-Sigmoid**: VGG19 image features, use 4 layers with 256 units each, use Side by Side Concatenation, use Sigmoid

4. Results

Table 1 shows the validation accuracy of our baseline model along with the individual methods, while table 2 shows the combined method accuracy.

As we can see from the tables and the plots [3, 4] we get a significant improvement (8.26%) from the baseline model if we use the method **4Layers256Units**. This method performs better in all three categories of answers i.e. *number*, *other* and *Yes/No*. Similar but slightly worse results are obtained by using the method **8Layers128Units**, which shows that increasing the depth of LSTM to be more than 4 layers is not beneficial. Surprisingly good results are obtained when using **ReLUAct** or any of the two fusion methods. On the other hand, methods like **AlexNet** and **SigmoidAct** offer negligible improvements. Using **ResNet50** offered modest boost in score by 3.13%.

Combining the best scoring methods to create a single model happens to not be the best design, i.e. "the best of all worlds" approach does not work. Actually, most of the individual methods perform noticeably better than when combined with another method.

5. Conclusion and Discussion

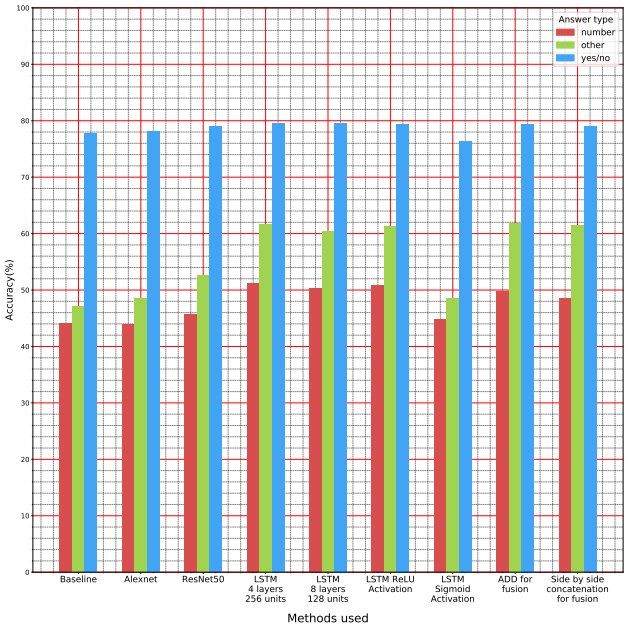
In this project we have provided extensive suite of experiments on how individual variations affect the score compared to the baseline model. We hope that these observations will be a strong foothold in solving VQA for abstract scenes and/or multiple choice questions. The general takeaway is that introducing changes to a single component of the entire model might drastically boost the performance, while finding a favorable combination of these variations remains a challenge. Through empirical evidence we have concluded that **4Layers256Units** method performs the best

Methods	Validation Accuracy			
	Overall	Number	Other	Y/N
Baseline	59.28	44.17	47.06	77.82
ResNet50	62.41	45.69	52.56	78.93
AlexNet	59.97	43.98	48.45	78.05
4Layers256Units	67.54	51.12	61.72	79.58
8Layers128Units	66.85	50.34	60.4	79.58
ReLUAct	67.24	50.75	61.29	79.42
SigmoidAct	59.47	44.76	48.6	76.4
ConcatSideBySide	66.85	48.46	61.53	79
ADDforFusion	67.37	49.76	61.9	79.42

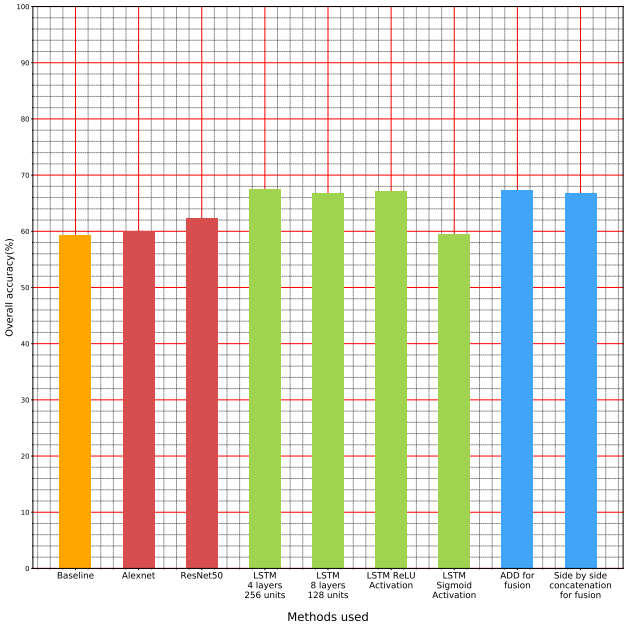
Table 1: Per answer type and overall validation accuracy for each individual method. All of the variants give improvement over the baseline model.

among all the variants and their combinations studied in this paper. The success of this particular setup might be related to the idea of breadth vs depth of neural networks. Using more layers enables us to acquire high level information about the question, but might introduce the vanishing gradient problem. On the other hand, using more units per layer is helpful for extracting low level details, but it is prone to over-fitting. We believe that the **4Layers256Units** method provides the best middle-ground for this trade-off, as our results confirm.

The task of VQA remains a challenge due to its multi-modal nature. Overcoming the language prior and creating balanced datasets are crucial for making progress. We believe that thorough experimentation and keeping an open mind about trying variations of existing well-performing models might positively surprise us or at least teach us valuable lessons, but most importantly bring us closer to solving VQA.



(a) Per question type accuracy for each variant



(b) Overall accuracy for each variant

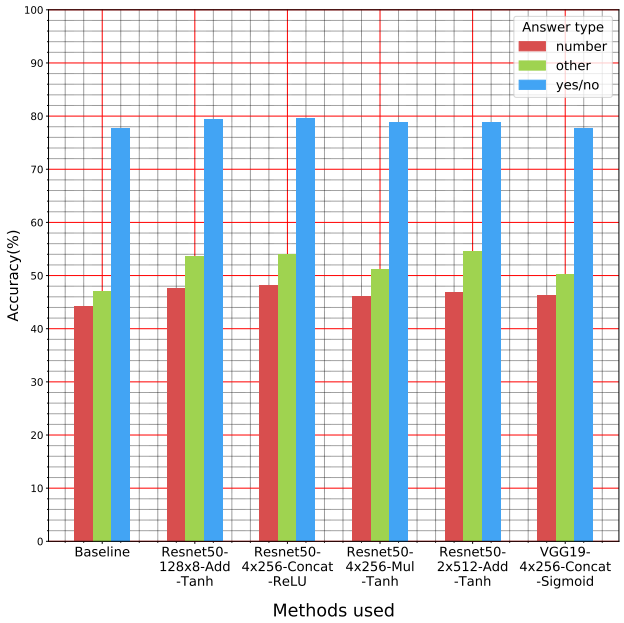
Figure 3: Validation accuracy for each variant. **4Layers156Units** outperforms the baseline for 8.26%

References

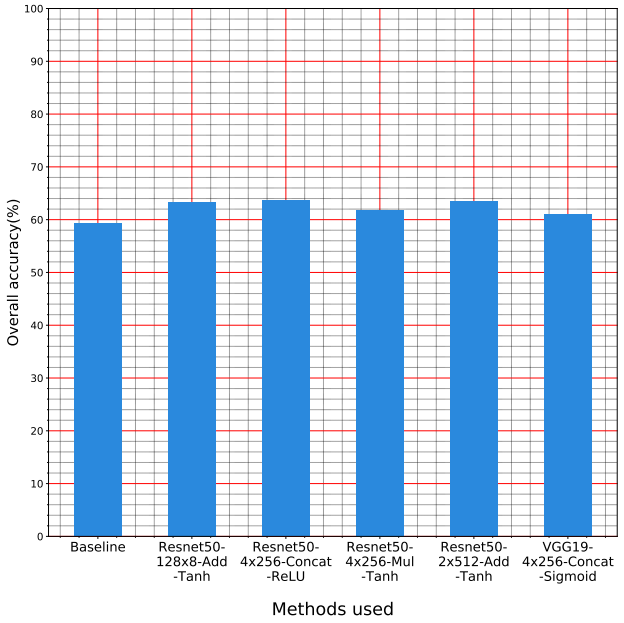
[1] Stanislaw Antol, Aishwarya Agrawal, Jiasen Lu, Margaret Mitchell, Dhruv Batra, C. Lawrence Zitnick, and Devi Parikh. VQA: Visual Question Answering. In *International Conference on Computer Vision (ICCV)*, 2015.

[2] Yash Goyal, Tejas Khot, Douglas Summers-Stay, Dhruv Batra, and Devi Parikh. Making the V in VQA matter: Elevating the role of image understanding in Visual Question Answering. In *Conference on Computer Vision and Pattern Recognition (CVPR)*, 2017.

[3] Peng Zhang, Yash Goyal, Douglas Summers-Stay, Dhruv Batra, and Devi Parikh. Yin and Yang: Balancing and answering binary visual questions. In *Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016.



(a) Per answer type validation accuracy for combined methods



(b) Overall accuracy for combined methods

Figure 4: Validation accuracy for combined methods

Combined method	Validation Accuracy			
	Overall	Number	Other	Y/N
ResNet50-2x512 -Add-Tanh	63.44	46.96	54.55	78.83
ResNet50-4x256 -Mul-Tanh	61.85	46.17	51.16	78.93
ResNet50-4x256 -Concat-ReLU	63.63	48.14	53.94	79.55
ResNet50-8x128 -Add-Tanh	63.32	47.57	53.6	79.37
VGG19-4x256-Concat-Sigmoid	61	46.34	50.27	77.76

Table 2: Per answer type and overall validation accuracy for each combined method. They all perform better than the baseline, but not as good as most individual variants.

[4] Jiasen Lu, Jianwei Yang, Dhruv Batra, and Devi Parikh. Hierarchical question-image co-attention for visual question answering, 2016.

[5] Kevin J. Shih, Saurabh Singh, and Derek Hoiem. Where to look: Focus regions for visual question answering. In *Computer Vision and Pattern Recognition*, 2016.

[6] Jiasen Lu, Xiao Lin, Dhruv Batra, and Devi Parikh. Deeper lstm and normalized cnn visual question answering model. https://github.com/VT-vision-lab/VQA_LSTM_CNN, 2015.

[7] Karen Simonyan and Andrew Zisserman. Very deep convolutional networks for large-scale image recognition, 2014.

[8] Alex Krizhevsky, Ilya Sutskever, and Geoffrey E. Hinton. Imagenet classification with deep convolutional neural networks. In *Advances in Neural Information Processing Systems*, page 2012.

[9] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. *arXiv preprint arXiv:1512.03385*, 2015.